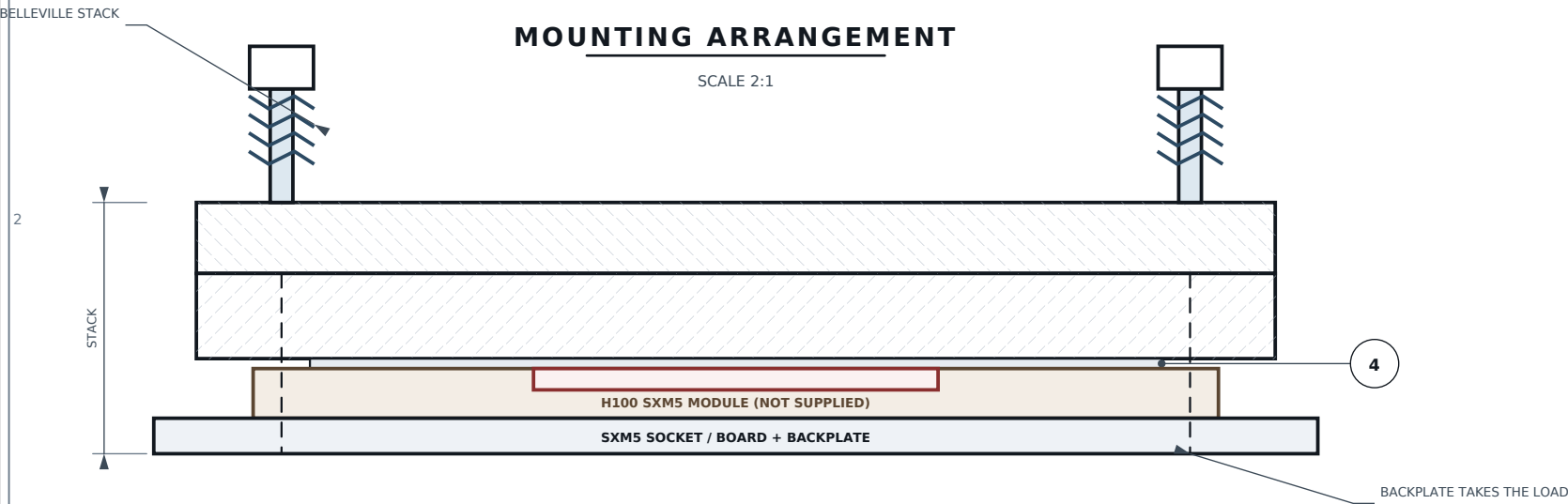


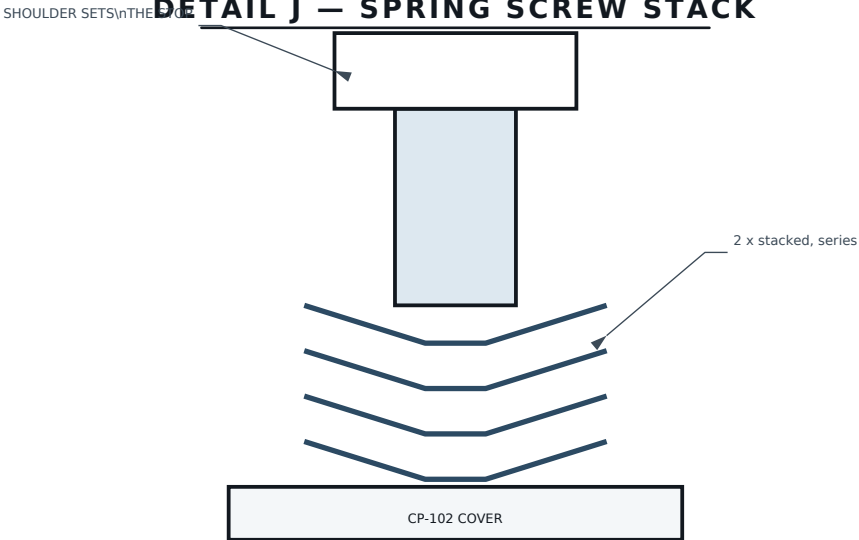
ASSUMED INTERFACE — CONFIRM EVERY DIMENSION AND THE PRELOAD AGAINST NVIDIA'S SXM5 MECHANICAL SPECIFICATION BEFORE MANUFACTURE

MOUNTING ARRANGEMENT

SCALE 2:1



DETAIL J — SPRING SCREW STACK



The shoulder, not the torque wrench, sets the preload.

WHY THIS SHEET IS NOT FINISHED

Torque-only assembly on a bare die cracks silicon.

- An SXM5 module is a bare die on a socketed carrier. The socket, the load frame and the allowable preload are defined by NVIDIA, not by the cold plate designer.
- Too much preload cracks the die or the substrate. Too little starves the TIM bond line and the 4.0 K TIM budget on sheet 1 becomes 15 K or worse.
- Obtain the SXM5 mechanical and thermal design guide, then set the preload, the screw shoulder height and the Belleville stack to match it. Everything else on these six sheets is independent of that number.
- Do not substitute a torque specification for a shoulder stop. Torque on an M3 screw scatters by +/- 30 % with friction alone.

MOUNTING PARAMETERS

PARAMETER	VALUE	SOURCE
Total preload on the package	TO BE TAKEN FROM NVIDIA SPEC	NOT SET HERE
4 Screws	4 x M3 x 0.5, shouldered	this drawing
Spring element	2 x stacked, series	this drawing
Tightening	cross pattern, 3 passes to shoulder	this drawing
Cold plate flatness, die face	0.015 max	CP-101
Parallelism, plate to package	0.05 max	this drawing
TIM	PTM, k >= 13 W/mK, 60 um BLT	CP-104

MOUNTING RISK REGISTER

RISK	MECHANISM	CONTROL
Die crack	preload above the package rating, or uneven load	shoulder stop + cross tightening
High TIM resistance	insufficient preload, or air in the bond line	preload to spec, controlled dispense
6 TIM pump-out	thermal cycling with a soft grease	use a phase-change material
Plate rocking	flatness or parallelism out of tolerance	0.015 / 0.05 per this sheet

INDIA-VC THERMAL HARDWARE

ISO 128 / ISO 2768-mK

TITLE

SXM5 MOUNTING INTERFACE

LOAD FRAME, PRELOAD AND TIM / DIMENSIONS ASSUMED — SEE BANNER

SCALE

2:1

UNITS

mm

THIRD ANGLE PROJECTION

DRAWING No.

CP-200

REV

A

SHEET

5 / 6

DRAWN

AI-ASSISTED DRAFT

CHECKED

— PENDING —

MATERIAL

SEE BOM

FINISH

SEE NOTES

DATE

2026-09-10